Module 2

Lab - Lists and Tuples

Tuples in Python

Code:
```
tuple1 = ("disco",10,1.2 )
tuple1
```

Output:
```
('disco', 10, 1.2)
```

Code:
```
type(tuple1)
```

Output:
```
<class 'tuple'>
```

Indexing

Code:
```
print(tuple1[0])
print(tuple1[1])
print(tuple1[2])
```

Output:
```
disco
10
1.2
```

Code:
```
print(type(tuple1[0]))
print(type(tuple1[1]))
print(type(tuple1[2]))
```

Output:
```
<class 'str'>
<class 'int'>
<class 'float'>
```

Code:
```
tuple1[-1]
```

Output:
```
1.2
```

Code:
```
tuple1[-2]
```

Output:
```
10
```

Code:
```
tuple1[-3]
```

Output:
```
'disco'
```

Concatenate Tuples

Code:
```
tuple2 = tuple1 + ("hard rock", 10)
tuple2
```

Output:
```
('disco', 10, 1.2, 'hard rock', 10)
```

Slicing

Code:
```
tuple2[0:3]
```

Output:
```
('disco', 10, 1.2)
```

Code:
```
tuple2[3:5]
```

Output:
```
('hard rock', 10)
```

Code:
```
len(tuple2)
```

Output:
```
5
```

Sorting

Code:
```
Ratings = (0, 9, 6, 5, 10, 8, 9, 6, 2)
```

Output:
```
no output, this line just stores the value
```

Code:
```
RatingsSorted = sorted(Ratings)
RatingsSorted
```

Output:
```
[0, 2, 5, 6, 6, 8, 9, 9, 10]
```

Nested Tuple

Code:
```
NestedT = (1, 2, ("pop", "rock") ,(3,4),("disco",(1,2)))
```

Output:
no output, this line just stores the value

Code:
```
print("Element 0 of Tuple: ", NestedT[0])
print("Element 1 of Tuple: ", NestedT[1])
print("Element 2 of Tuple: ", NestedT[2])
print("Element 3 of Tuple: ", NestedT[3])
print("Element 4 of Tuple: ", NestedT[4])
```

Output:
```
Element 0 of Tuple:  1
Element 1 of Tuple:  2
Element 2 of Tuple:  ('pop', 'rock')
Element 3 of Tuple:  (3, 4)
Element 4 of Tuple:  ('disco', (1, 2))
```

Code:
```
print("Element 2, 0 of Tuple: ",   NestedT[2][0])
print("Element 2, 1 of Tuple: ",   NestedT[2][1])
print("Element 3, 0 of Tuple: ",   NestedT[3][0])
print("Element 3, 1 of Tuple: ",   NestedT[3][1])
print("Element 4, 0 of Tuple: ",   NestedT[4][0])
print("Element 4, 1 of Tuple: ",   NestedT[4][1])
```

Output:
```
Element 2, 0 of Tuple:  pop
Element 2, 1 of Tuple:  rock
Element 3, 0 of Tuple:  3
Element 3, 1 of Tuple:  4
Element 4, 0 of Tuple:  disco
Element 4, 1 of Tuple:  (1, 2)
```

Code:
```
NestedT[2][1][0]
```

Output:
```
'r'
```

Code:
```
NestedT[2][1][1]
```

Output:
```
'o'
```

Code:
```
NestedT[4][1][0]
```

Output:
```
1
```

Code:
```
NestedT[4][1][1]
```

Output:
```
2
```

Quiz on Tuples

Code:
```
genres_tuple = ("pop", "rock", "soul", "hard rock", "soft rock", \
                "R&B", "progressive rock", "disco")
genres_tuple
```

Output:
```
('pop', 'rock', 'soul', 'hard rock', 'soft rock', 'R&B', 'progressive rock',
'disco')
```

Code:
```
len(genres_tuple)
```

Output:
```
8
```

Code:
```
genres_tuple[3]
```

Output:
```
'hard rock'
```

Code:
```
genres_tuple[3:6]
```

Output:
```
('hard rock', 'soft rock', 'R&B')
```

Code:
```
genres_tuple[0:2]
```

Output:
```
('pop', 'rock')
```

Code:
```
"disco".find('s')
```

Output:
```
2
```

Code:
```
C_tuple = (-5, 1, -3)
C_list = sorted(C_tuple)
C_list
```

Output:

```
[-5, -3, 1]
```

```
Lab completed - Tuples in Python.
```

Lists in Python

```
Indexing
```

Code:
```
L = ["The Bodyguard", 7.0, 1992]
L
```

Output:
```
['The Bodyguard', 7.0, 1992]
```

Code:
```
print('the same element using negative and positive indexing:\n
Postive:',L[0],
'\n Negative:' , L[-3]  )
print('the same element using negative and positive indexing:\n
Postive:',L[1],
'\n Negative:' , L[-2]  )
print('the same element using negative and positive indexing:\n
Postive:',L[2],
'\n Negative:' , L[-1]  )
```

Output:
```
the same element using negative and positive indexing:
 Postive: The Bodyguard
 Negative: The Bodyguard
the same element using negative and positive indexing:
 Postive: 7.0
 Negative: 7.0
the same element using negative and positive indexing:
 Postive: 1992
 Negative: 1992
```

```
List Content
```

Code:
```
["The Bodyguard", 7.0, 1992, [1, 2], ("A", 1)]
```

Output:
```
['The Bodyguard', 7.0, 1992, [1, 2], ('A', 1)]
```

```
List Operations
```

Code:
```
L = ["The Bodyguard", 7.0,1992,"BG",1]
L
```

Output:
```
['The Bodyguard', 7.0, 1992, 'BG', 1]
```

Code:
```
L[3:5]
```

Output:
```
['BG', 1]
```

Code:
```
L = [ "The Bodyguard", 7.0]
L.extend(['pop', 10])
L
```

Output:
```
['The Bodyguard', 7.0, 'pop', 10]
```

Code:
```
L = [ "The Bodyguard", 7.0]
L.append(['pop', 10])
L
```

Output:
```
['The Bodyguard', 7.0, ['pop', 10]]
```

Code:
```
L.append(['a','b'])
L
```

Output:
```
['The Bodyguard', 7.0, ['pop', 10], ['a', 'b']]
```

Code:
```
A = ["disco", 10, 1.2]
print('Before change:', A)
A[0] = 'hard rock'
print('After change:', A)
```

Output:
```
Before change: ['disco', 10, 1.2]
After change: ['hard rock', 10, 1.2]
```

Code:
```
print('Before change:', A)
del(A[0])
print('After change:', A)
```

Output:
```
Before change: ['hard rock', 10, 1.2]
After change: [10, 1.2]
```

Code:
```
'hard rock'.split()
```

Output:
```
['hard', 'rock']
```

Code:
```
'A,B,C,D'.split(',')
```

Output:
```
['A', 'B', 'C', 'D']
```

Copy and Clone List

Code:
```
A = ["hard rock", 10, 1.2]
B = A
print('A:', A)
print('B:', B)
```

Output:
```
A: ['hard rock', 10, 1.2]
B: ['hard rock', 10, 1.2]
```

Code:
```
print('B[0]:', B[0])
A[0] = "banana"
print('B[0]:', B[0])
```

Output:
```
B[0]: hard rock
B[0]: banana
```

Code:
```
B = A[:]
B
```

Output:
```
['banana', 10, 1.2]
```

Code:
```
print('B[0]:', B[0])
A[0] = "hard rock"
print('B[0]:', B[0])
```

Output:
```
B[0]: banana
B[0]: banana
```

Quiz on List

Code:
```
a_list = [1, 'hello', [1,2,3] , True]
a_list
```

Output:

```
[1, 'hello', [1, 2, 3], True]
```

Code:
```
a_list[1]
```

Output:
```
'hello'
```

Code:
```
a_list[1:4]
```

Output:
```
['hello', [1, 2, 3], True]
```

Code:
```
A = [1, 'a']
B = [2, 1, 'd']
A + B
```

Output:
```
[1, 'a', 2, 1, 'd']
```

Scenario: Shopping List

Task-1 Create an empty list

Code:
```
Shopping_list = []
Shopping_list
```

Output:
```
[]
```

Task-2 Now store the number of items to the shopping_list

Code:
```
Shopping_list = ["Watch", "Laptop", "Shoes", "Pen", "Clothes"]
Shopping_list
```

Output:
```
['Watch', 'Laptop', 'Shoes', 'Pen', 'Clothes']
```

Task-3 Add a new item to the shopping_list

Code:
```
Shopping_list.append("Football")
Shopping_list
```

Output:
```
['Watch', 'Laptop', 'Shoes', 'Pen', 'Clothes', 'Football']
```

Task-4 Print First item from the shopping_list

Code:

```
print(Shopping_list[0])
```

Output:
```
Watch
```

Task-5 Print Last item from the shopping_list

Code:
```
print(Shopping_list[-1])
```

Output:
```
Football
```

Task-6 Print the entire Shopping List

Code:
```
print(Shopping_list)
```

Output:
```
['Watch', 'Laptop', 'Shoes', 'Pen', 'Clothes', 'Football']
```

Task-7 Print the items that are important to buy from the Shopping List

Code:
```
print(Shopping_list[1:3])
```

Output:
```
['Laptop', 'Shoes']
```

Task-8 Change the item from the shopping_list

Code:
```
Shopping_list[3] = "Notebook"
Shopping_list
```

Output:
```
['Watch', 'Laptop', 'Shoes', 'Notebook', 'Clothes', 'Football']
```

Task-9 Delete the item from the shopping_list that is not required

Code:
```
del(Shopping_list[4])
Shopping_list
```

Output:
```
['Watch', 'Laptop', 'Shoes', 'Notebook', 'Football']
```

Task-10 Print the shopping list

Code:
```
print(Shopping_list)
```

Output:
```
['Watch', 'Laptop', 'Shoes', 'Notebook', 'Football']
```

Lab completed - Lists in Python.

Sets

Code:

```
set1 = {"pop", "rock", "soul", "hard rock", "rock", "R&B", "rock", "disco"}
set1
```

output:

```
{'R&B', 'disco', 'hard rock', 'pop', 'rock', 'soul'}
```

Code:

```
album_list = [ "Michael Jackson", "Thriller", 1982, "00:42:19", \
                    "Pop, Rock, R&B", 46.0, 65, "30-Nov-82", None, 10.0]
album_set = set(album_list)
album_set
```

output:

```
{'00:42:19',
 10.0,
 1982,
 '30-Nov-82',
 46.0,
 65,
 'Michael Jackson',
 None,
 'Pop, Rock, R&B',
 'Thriller'}
```

Code:

```
music_genres = set(["pop", "pop", "rock", "folk rock", "hard rock", "soul", \
                "progressive rock", "soft rock", "R&B", "disco"])
music_genres
```

output:

```
{'R&B',
 'disco',
 'folk rock',
 'hard rock',
 'pop',
 'progressive rock',
 'rock',
 'soft rock',
 'soul'}
```

Code:

```
A = set(["Thriller", "Back in Black", "AC/DC"])
A
```

Output:

```
{'AC/DC', 'Back in Black', 'Thriller'}
```

Code:

```
A.add("NSYNC")
A
```

Ouput:

```
{'AC/DC', 'Back in Black', 'NSYNC', 'Thriller'}
```

Code:

```
A.add("NSYNC")
A
```

Output:

```
{'AC/DC', 'Back in Black', 'NSYNC', 'Thriller'}
```

Code:

```
A.remove("NSYNC")
A
```

Output:

```
{'AC/DC', 'Back in Black', 'Thriller'}
```

Code:

```
"AC/DC" in A
```

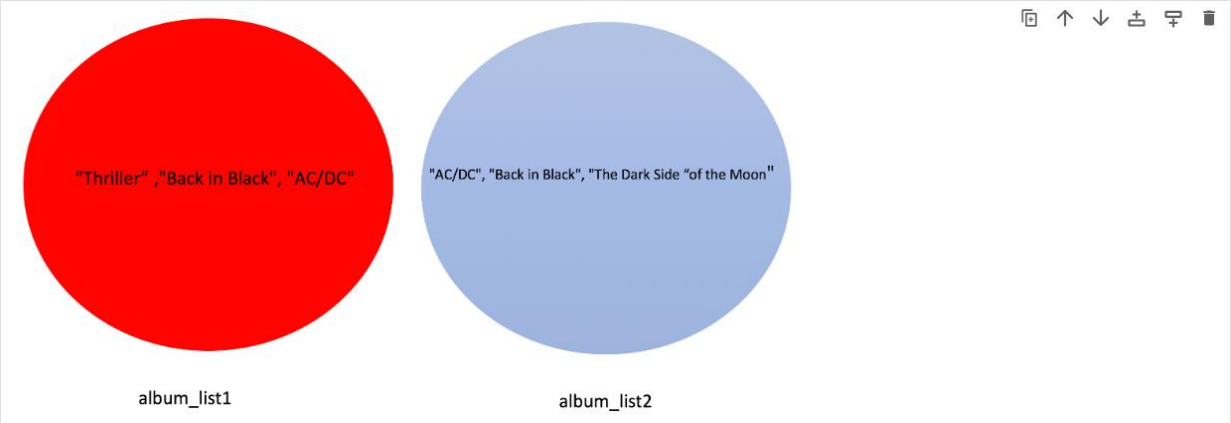
Output:

```
True
```

Code:

```
album_set1 = set(["Thriller", 'AC/DC', 'Back in Black'])
album_set2 = set([ "AC/DC", "Back in Black", "The Dark Side of the Moon"])
```

output:



Code:

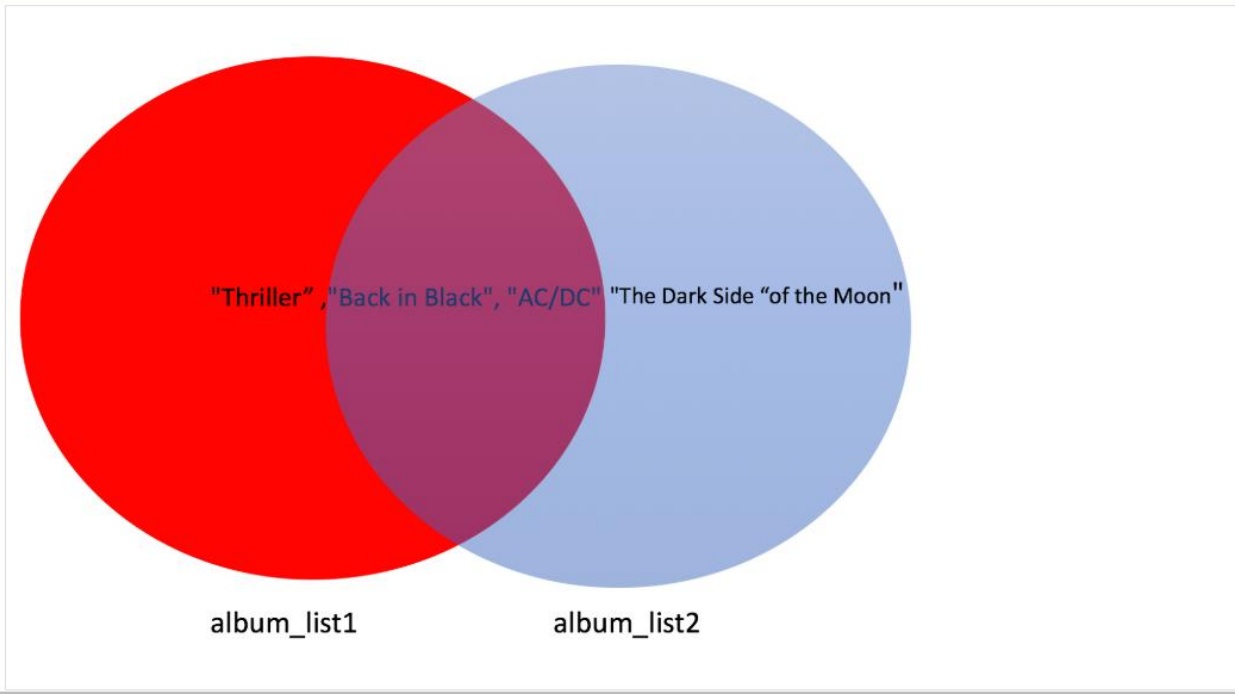
album_set1, album_set2

output:

({'AC/DC', 'Back in Black', 'Thriller'},

{'AC/DC', 'Back in Black', 'The Dark Side of the Moon'})

As both sets contain **AC/DC** and **Back in Black** we represent these common elements with the intersection of two circles.



Code:

intersection = album_set1 & album_set2

intersection

output:

{'AC/DC', 'Back in Black'}

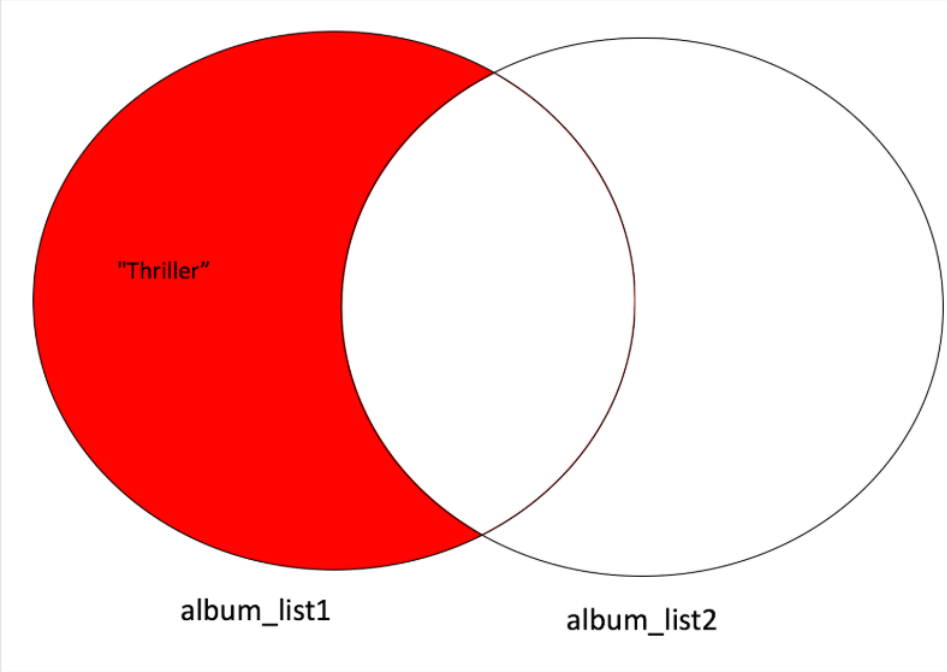
Code:

album_set1.difference(album_set2)

output:

{'Thriller'}

You only need to consider elements in `album_set1`; all the elements in `album_set2`, including the intersection, are not included.



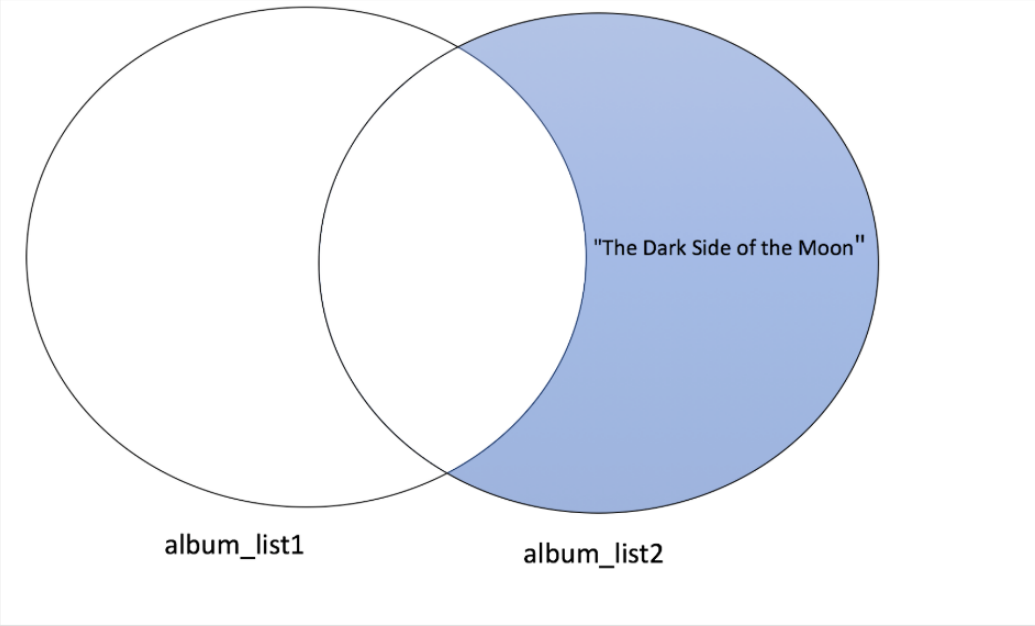
The elements in `album_set2` but not in `album_set1` is given by:

Code:

album_set2.difference(album_set1)

output:

{'The Dark Side of the Moon'}



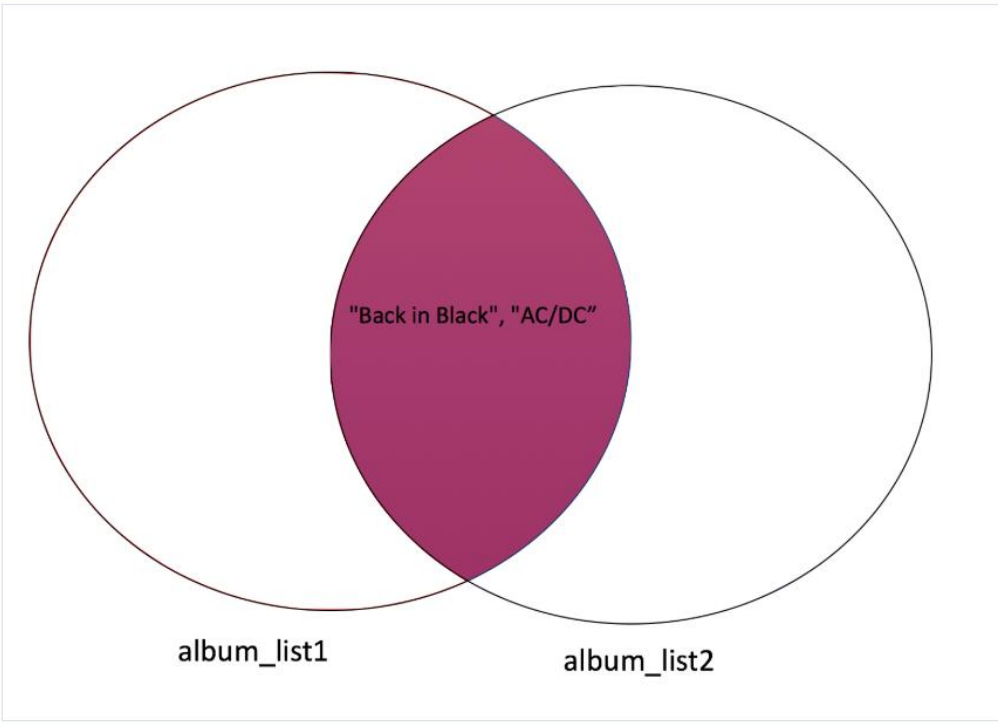
You can also find the intersection of `album_list1` and `album_list2`, using the `intersection` method:

Code:

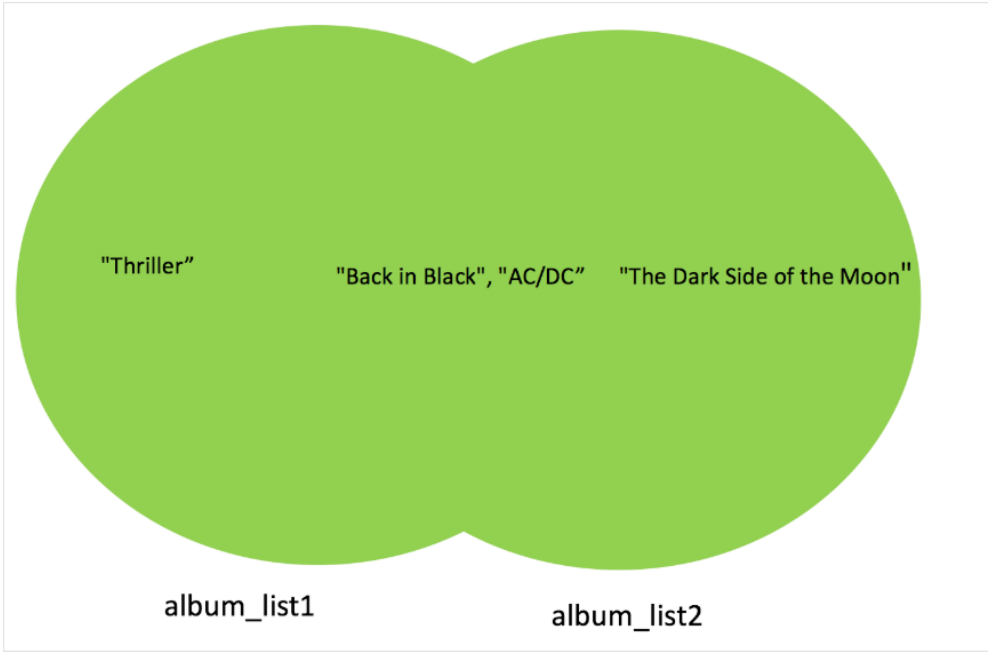
album_set1.intersection(album_set2)

output:

{'AC/DC', 'Back in Black'}



The union corresponds to all the elements in both sets, which is represented by coloring both circles:



Code:

```
album_set1.union(album_set2)
```

output:

{'AC/DC', 'Back in Black', 'The Dark Side of the Moon', 'Thriller'}

Code:

```
set(album_set1).issuperset(album_set2)
```

output:

False

Code:

```
set(album_set2).issubset(album_set1)
```

output:

False

Code:

```
set({"Back in Black", "AC/DC"}).issubset(album_set1)
```

output:

True

Code:

```
album_set1.issuperset({"Back in Black", "AC/DC"})
```

output:

True

Quiz on Sets

Code:

```
album_list = ['rap', 'house', 'electronic music', 'rap']

album_set = set(album_list)

album_set
```

output:

set(['rap','house','electronic music','rap'])

code:

```
A = [1, 2, 2, 1]
B = set([1, 2, 2, 1])
print("sum(A) =", sum(A))
print("sum(B) =", sum(B))
print("sum(A) == sum(B) =", sum(A) == sum(B))
```

output:

sum(A) = 6

sum(B) = 3

sum(A) == sum(B) = False

code:

```
album_set1 = set(["Thriller", "AC/DC", "Back in Black"])
album_set2 = set(["AC/DC", "Back in Black", "The Dark Side of the Moon"])
album_set3 = album_set1.union(album_set2)
print("album_set3 =", album_set3)
```

output:

album_set3 = {'Thriller', 'AC/DC', 'Back in Black', 'The Dark Side of the Moon'}

code:

```
album_set1 = set(["Thriller", "AC/DC", "Back in Black"])
album_set2 = set(["AC/DC", "Back in Black", "The Dark Side of the Moon"])
album_set3 = album_set1.union(album_set2)
print("Is album_set1 a subset of album_set3?")
print(album_set1.issubset(album_set3))
```

output:

Is album_set1 a subset of album_set3?

True

Lab – Dictionaries

Code:

```
Dict = {"key1": 1, "key2": "2", "key3": [3, 3, 3], "key4": (4, 4, 4), ('key5'): 5, (0, 1): 6}

Dict
```

Output:

```
{'key1': 1,

 'key2': '2',

 'key3': [3, 3, 3],

 'key4': (4, 4, 4),

 'key5': 5,

 (0, 1): 6}
```

Code:

```
Dict["key1"]
```

Output:

1

Code:

```
Dict[(0, 1)]
```

Output:

6

Code:

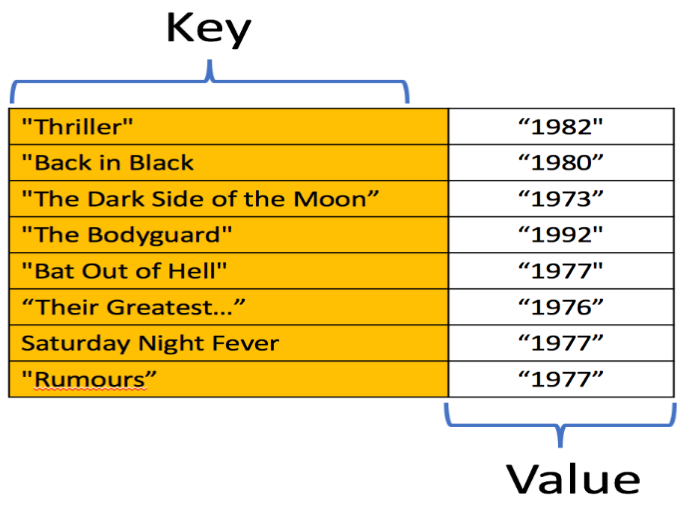
```
release_year_dict = {"Thriller": "1982", "Back in Black": "1980", \
                     "The Dark Side of the Moon": "1973", "The Bodyguard": "1992", \
                     "Bat Out of Hell": "1977", "Their Greatest Hits (1971-1975)": "1976", \
                     "Saturday Night Fever": "1977", "Rumours": "1977"}
release_year_dict
```

output:

```
{'Thriller': '1982',
 'Back in Black': '1980',
 'The Dark Side of the Moon': '1973',
 'The Bodyguard': '1992',
 'Bat Out of Hell': '1977',
 'Their Greatest Hits (1971-1975)': '1976',
 'Saturday Night Fever': '1977',
 'Rumours': '1977'}
```

Keys

It is helpful to visualize the dictionary as a table, as in the following image. The first column represents the keys, the second column represents the values.



Code:

release_year_dict['Thriller']

output:

‘1982'

This corresponds to:

"Thriller"	“1982"
"Back in Black	“1980”
"The Dark Side of the Moon”	“1973”
"The Bodyguard"	“1992"
"Bat Out of Hell"	“1977"
“Their Greatest…”	“1976”
“Saturday Night Fever”	“1977”
"Rumours”	“1977”

Code:

release_year_dict['The Bodyguard']

output:

'1992'

"Thriller"	“1982"
"Back in Black	“1980”
"The Dark Side of the Moon”	“1973”
"The Bodyguard"	“1992"
"Bat Out of Hell"	“1977"
“Their Greatest…”	“1976”
“Saturday Night Fever”	“1977”
"Rumours”	“1977”

Now let us retrieve the keys of the dictionary using the method `keys()`:

Code:

release_year_dict.keys()

output:

dict_keys(['Thriller', 'Back in Black', 'The Dark Side of the Moon', 'The Bodyguard', 'Bat Out of Hell', 'Their Greatest Hits (1971-1975)', 'Saturday Night Fever', 'Rumours'])

code:

release_year_dict.values()

output:

dict_values(['1982', '1980', '1973', '1992', '1977', '1976', '1977', '1977'])

code:

release_year_dict['Graduation'] = '2007'

release_year_dict

output:

{ 'Thriller': '1982',

 'Back in Black': '1980',

 'The Dark Side of the Moon': '1973',

 'The Bodyguard': '1992',

 'Bat Out of Hell': '1977',

 'Their Greatest Hits (1971-1975)': '1976',

 'Saturday Night Fever': '1977',

 'Rumours': '1977',

 'Graduation': '2007'}

Code:

```
del(release_year_dict['Thriller'])

del(release_year_dict['Graduation'])

release_year_dict
```

output:

{'Back in Black': '1980',

 'The Dark Side of the Moon': '1973',

 'The Bodyguard': '1992',

 'Bat Out of Hell': '1977',

 'Their Greatest Hits (1971-1975)': '1976',

 'Saturday Night Fever': '1977',

 'Rumours': '1977'}

Code:

'The Bodyguard' in release_year_dict

Output:

True

Quiz on Dictionaries

Code:

```
album_sales_dict = {
   "The Bodyguard": 50,
   "Back in Black": 50,
   "Thriller": 65
}
print(album_sales_dict)
```

output:

{'The Bodyguard': 50, 'Back in Black': 50, 'Thriller': 65}

Code:

```
album_sales_dict = {
   "The Bodyguard": 50,
   "Back in Black": 50,
   "Thriller": 65
}
print(album_sales_dict["Thriller"])
```

output:

65

Code:

```
album_sales_dict = {
    "The Bodyguard": 50,
    "Back in Black": 50,
    "Thriller": 65
}
print(album_sales_dict.keys())
```

output:

dict_keys(['The Bodyguard', 'Back in Black', 'Thriller'])

code:

```
album_sales_dict = {
    "The Bodyguard": 50,
    "Back in Black": 50,
    "Thriller": 65
}
print(album_sales_dict.values())
```

output:

dict_values([50, 50, 65])

code:

```
soundtrack_dic = {
    "The Bodyguard": "1992",
    "Saturday Night Fever": "1977"
}
print(soundtrack_dic.keys())
```

output:

dict_keys(['The Bodyguard', 'Saturday Night Fever'])

code:

```
soundtrack_dic = {
   "The Bodyguard": "1992",
   "Saturday Night Fever": "1977"
}
print(soundtrack_dic.values())
```

output:

dict_values(['1992', '1977'])